Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User requests to find all tall buildings in Exeter over 18m or 7 floors	human, host_agent	Message from human to host_agent: "Find all tall buildings in Exeter that are over 18m tall or 7 floors high"	User initiates the query.
2	Plan the solution steps for the geospatial query	host_agent, planning_agent	send_message to planning_agent with task_description: "User wants to find all tall buildings in Exeter that are over 18m tall or 7 floors high. Provide the general steps to solve this geospatial query."	planning_agent is chosen to break down the query into actionable steps as per its prompt.
3	Planning agent decomposes the query	planning_agent	Output steps: ["Find Exeter 1 area", "Search for tall buildings in Exeter as many search results", "Apply condition: buildings over 18m tall or 7 floors high"]	Steps are based on the planning_agent's prompt and reasoning template for geospatial queries.
4	Find the area polygon for Exeter	host_agent, named_area_agent	send_message to named_area_agent with task_description: "Find the area polygon for Exeter."	named_area_agent is chosen because the prompt specifies it handles named area polygons for cities like Exeter.
5	Search for Exeter polygon in OS NGD	named_area_agent	call_os_ngd with polygon_or_point=True, filename='exeter_polygon_data'	polygon_or_point=True is set to get area polygons; filename is descriptive for the artifact.
6	Filter the polygon data to get only Exeter	host_agent, data_analysis_agent	send_message to data_analysis_agent: "Filter the artifact 'exeter_polygon_data' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon_data'."	Filtering is needed because the named area search returns multiple polygons; data_analysis_agent is used for precise filtering.
7	Generate metadata for 'exeter_polygon_data'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['exeter_polygon_data']	Metadata is generated to understand the schema and columns for accurate filtering.
8	Filter for Exeter polygon using code	data_analysis_agent, code_executor	code_executor with code that filters for name1_text == 'exeter' and relevant descriptiongroup/description values; artifact_names=['exeter_polygon_data']	name1_text and descriptiongroup/description columns are used based on metadata and the need to identify the city polygon.
9	Confirm filtered Exeter polygon artifact	data_analysis_agent	Output: 1 matching polygon saved as 'filtered_exeter_polygon_data'	Confirms that the correct polygon for Exeter has been isolated.
10	Search for buildings in Exeter polygon	host_agent, buildings_agent	send_message to buildings_agent: "Search for buildings in the area defined by the artifact 'filtered_exeter_polygon_data' that are over 18m tall or have 7 or more floors."	buildings_agent is chosen as per its prompt for building searches; bbox is set to the filtered Exeter polygon.
11	Query OS NGD for buildings in Exeter	buildings_agent	call_os_ngd with bbox='filtered_exeter_polygon_data', polygon_or_point=True, filename='exeter_all_buildings_for_height_floor_filter'	bbox parameter is set to the Exeter polygon artifact; polygon_or_point=True for building polygons.
12	Filter buildings by height/floors	host_agent, data_analysis_agent	send_message to data_analysis_agent: "From the artifact 'buildings_exeter_all_buildings_for_height_floor_filter', filter and return only those buildings that are either over 18m tall (using the 'relativeroofbase' column for building height) or have 7 or more floors (using the 'building_numfloors' column). Save the filtered results as 'exeter_buildings_over_18m_or_7floors'."	data_analysis_agent is used to apply the final conditions; columns are chosen based on metadata and buildings_agent prompt.
13	Generate metadata for buildings artifact	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['buildings_exeter_all_buildings_for_height_floor_filter']	Ensures correct column names and types for filtering.
14	Filter buildings using code	data_analysis_agent, code_executor	code_executor with code filtering (height_relativeroofbase_m > 18) or (numberoffloors >= 7); artifact_names= ['buildings_exeter_all_buildings_for_height_floor_filter']	Columns are chosen based on metadata and buildings_agent prompt; code applies the OR condition as per user query.
15	Confirm filtered tall buildings artifact	data_analysis_agent	Output: 55 buildings in Exeter meet the criteria, saved as 'exeter_buildings_over_18m_or_7floors'	Confirms the filtering step and artifact creation.
16	Plot the tall buildings on a map	host_agent, plotting_agent	send_message to plotting_agent: "Plot the buildings from the artifact 'exeter_buildings_over_18m_or_7floors' to show all tall buildings in Exeter that are over 18m tall or have 7 or more floors."	plotting_agent is chosen for visualization; artifact name is passed for plotting.
17	Generate metadata for plotting	plotting_agent	generate_metadata_for_artifacts with artifact_names=['exeter_buildings_over_18m_or_7floors']	Ensures correct columns and CRS for plotting.
18	Plot buildings using code	plotting_agent, code_executor	code_executor with code to plot polygons on a folium map, save as 'exeter_tall_buildings_map.html'; artifact_names=['exeter_buildings_over_18m_or_7floors']	Code uses folium for interactive mapping; columns for description, height, and floors are used for popups.
19	Confirm map artifact and summary	plotting_agent	Output: Map with 55 tall buildings, artifact 'exeter_tall_buildings_map', summary of first 5 buildings	Confirms successful visualization and artifact creation.
20	Final summary to user	host_agent	Output: All tall buildings in Exeter over 18m or 7+ floors identified and mapped; 55 buildings found and plotted	host_agent summarizes the entire workflow and results for the user.

Overall Summary: - Databases used: Ordnance Survey NGD for named areas and buildings. - Filters/code: Initial polygon search for Exeter, then filtering for the city polygon using name and description columns. Buildings were searched within this polygon, and then filtered by height (height_relativeroofbase_m > 18) or number of floors (numberoffloors >= 7) using code. Visualization was done using folium. - Final outcome: 55 buildings in Exeter were found that are either over 18m tall or have 7 or more floors. These were plotted on an interactive map (artifact: exeter_tall_buildings_map.html). - Overall reasoning: Each agent and tool was chosen based on its specialized prompt and capabilities. Metadata was used to ensure correct column usage. Filtering and plotting steps were performed using code, with careful attention to the schema and user requirements. The workflow ensured that only relevant buildings in the correct area were included and visualized.